وزارة التربية والتعليم والتعليم الفنى
وحدة تشغيل و إدارة مدارس التكنولوجيا التطبيقية
مدرسة we للتكنولوجيا التطبيقية

1)complete each of the following :

Indefinite integral and trigonometric functions.	
1- $\int 15\ x^4\,dx =$	
2- $\int x^7\,dx$=	
3- $\int x^6\,dx =$	
4- $\int \frac{dx}{x^4}$	
5- $\int 4\,dx$	
6- $\int 2\,dx$	
7- $\int \frac{1}{x}\,dx$	
8- $\int x^{\frac{1}{3}}\,dx$	
9- $\int x^{\frac{1}{4}}\,dx$ =	
10- $\int \left(\frac{3}{x}+\frac{5}{x^2}\right)dx =$	
11- $\int \left(x^2-x+2\right)dx$	
12- $\int \left(x^2+x-1\right)dx$	
13- $\int \frac{-7}{\sqrt[3]{x^2}}\,dx$	
14- $\int \left(3t^2-4t+7\right)dt =$	
15- $\int \left(2t^2+5t-3\right)dt$	
$16 - \int 2e^{2x}\,dx$ =	

17-$\int \quad 6e^{8x}\,dx$	
18- $\int \quad (x+4)^2\,dx = \int \quad x^2+8x+16$ dx	
19-$\int \quad \frac{5}{\sqrt[4]{x^3}}\,dx$	
20-$\int \quad \frac{2}{3}\,e^{-9x}dx$	
21-$\int \quad \left(\frac{3}{x}-5e^{2x}+\sqrt{x^7}\right)\,dx$	
22-$\int \quad 4e^{4x}dx$	
23-$\int \quad \left(5x^2-2e^{7x}\right)dx$	
$24-\int \quad e^{5x}\,dx$	
25-$\int \quad e^{6x}\,dx$	

Answer

1	$3x^5+c$
2	$\frac{x^8}{8}+c$
3	$\frac{x^7}{7}+c$
4	- $\frac{1}{3x^3}+c$
5	$4x+c$
6	2x+c
7	$ln\,x+c$
8	$\frac{3}{4}x^{\frac{4}{3}}+c$
9	$\frac{4}{5}x^{\frac{5}{4}}$ +c
10	$3ln\,x-\frac{5}{x}+c$

11	$\frac{x^3}{3}-\frac{x^2}{2}+2x+c$
12	$\frac{x^3}{3}+\frac{x^2}{2}-x+c$
13	$-21x^{\frac{1}{3}}+c$
14	t^3-2t^2+7t+c
15	$t^3+\frac{5}{2}t^2-3t+c$
16	$e^{2x}+c$
17	$\frac{6}{8}e^{8x}+c$
18	$\frac{x^3}{3}+4x^2+16x+c$
19	$20\,x^{\frac{1}{4}}+c$
20	$\frac{-2}{27}e^{-9x}+c$
21	$3ln\,x-\frac{5}{2}\,e^{2x}+\frac{x^{\frac{9}{2}}}{\frac{9}{2}}+c$
22	$e^{4x}+c$
23	$\frac{5}{3}x^3-\frac{2}{7}e^{7x}+c$
24	$\frac{1}{5}e^{5x}+c$
25	$\frac{e^{6x}}{6}+c$

2) complete each of the following

Integration by parts.		
1	$\int\ 4x\,e^{4x}\,dx$	
2	$\int\ 3xe^{3x}\,dx$	
3	$\int\ xe^{5x}dx$	

4	$\int \quad 2xe^{4x}dx$	

answer

1	$4xe^{4x} - \frac{1}{4}e^{4x} + c$
2	$xe^{3x} - \frac{1}{3}e^{3x} + c$
3	$\frac{1}{5}xe^{5x} - \frac{1}{25}e^{5x}$+c
4	$\frac{1}{2}xe^{4x} - \frac{1}{8}e^{4x}$+c

3) complete each of the following

Definite integration		
1	$\int_{1}^{2} (x^2 - 3)\, dx$	
2	$\int_{0}^{\pi} cos\, x \;\; dx$	
3	$\int_{0}^{\frac{\pi}{4}} sec^2\, x \;\; dx$	
4	$\int_{0}^{5} 6x\, dx$	
5	$\int_{-3}^{1} 8 \;\; dt$	

answer

1	$\frac{-2}{3}$
2	0
3	1
4	75
5	32

3)calculate the area of the plane region bounded by:

Applications on integration (Area in the plane)		
1	a) The curve $y = 3$, X-axis and the two straight lines $x = 0$ and $x = 4$	
2	a) The curve $y = 4x$, X-axis and the two straight lines $x = -1$ and $x = 5$	
3	a) The curve $y = 3x^2$, X- axis and the two straight lines $x = -2$ and $x = 2$	

Answer

1	$$\int_0^4 3\,dx = [3x]_0^4 = 12$$
2	$$\int_{-1}^5 4x\,dx = [2x^2]_{-1}^5 = 48$$
3	$$\int_{-2}^2 3x^2\,dx = [x^3]_{-2}^2 = 16$$

1)Choose the correct answer:

	A	B	C	D
1) What is the mean in the following sample? 8, 1, 10, 4, 1, 1, 14, 26, 25	1	8	10	13.5
2) What is the mean in the following sample? 3, 16, 23, 14, 24, 21, 13, 24, 24	21	24	13.5	18
3) What is the median in the following sample? 2, 18, 22, 9, 22, 12, 22, 6, 22	18	22	15	12

4) What is the median in the following sample? 30, 15, 6, 4, 6, 6, 17, 27, 30, 27	27	16	15.5	30
5) What is the mode in the following sample? 2, 30, 3, 2, 30, 2, 9, 10	2	10	11	30
6) what the mean in the sample? 7, 29, 15, 4, 3, 3, 3, 5, 3, 2.	10	7	7.4	8

Answer

1	c-10
2	d-18
3	a)18
4	b)16
5	a-2
6	c)7.4

2) find

1	If the range and the smallest value of a set of data are 36 and 10 respectively, then the largest value is	
2	Ten percent of the tires produced by a tire company are defective. If two tires are selected at random. What is the probability that they are both good?	
3	If the probability of passing Math is 70%, the probability of passing English 80% and the probability of passing both Math and English is 60%, then find the probability of passing Math or English.	

answer

1	36+10=46
2	that the tire is good is $P(G) =$ $Since\ ten\ percent\ of\ the\ tires\ are\ defective$

	$P(D) = \frac{10}{100}$ $= \frac{1}{10} \, and \; thus, \; the \; probability 1$ $- \frac{1}{10} = \frac{9}{10}$ The probability that both tires are good is: $P(G) \times P(G) = \frac{9}{10} \times \frac{9}{10} = \frac{81}{100}$
3	The probability of passing Math or English $= P(M \; Or \; E) = P(M) + P(E) - P(M \; and \; E)$ $= 70\% + 80\% - 60\% = 90\%$